

Practice set – Color of complexes

Q1. Based on crystal field theory (CFT), write the electronic configuration of d^4 in terms of t_{2g} and e_g in an octahedral field given crystal field splitting energy (Δ_o) is greater than pairing energy (P).

Q2. Calculate the CFSE for the following complexes: $[\text{CoF}_6]^{3-}$, $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$, $[\text{Co}(\text{NH}_3)_6]^{3+}$, and $[\text{Co}(\text{en})_6]^{3+}$. Arrange them in the increasing order of CFSE. [Atomic number of Co is 27]

Q3. Predict what possible transition(s) do or do not contribute towards the intense blood-red color of iron thiocyanate $[\text{Fe}(\text{SCN})_3]$, formed during the Lassaigue's test for nitrogen and sulfur. [Atomic number of Fe is 26] [Detailed explanation is required]

Q4. Explain which of the following complexes is paramagnetic?

- (a) Hexamine chromium(III) ion (b) tetraammine zinc(II) ion
(c) tetracyanonickelate ion (d) diamine silver(I) ion

[Atomic number of Cr, Zn, Ni and Ag is 24, 30, 28 and 47 respectively]

Q5. Explain the violet color of $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$ complex on the basis of crystal field theory. [Atomic number of Ti is 22]

Q6. What are the factors that affect the crystal field splitting energy. Describe with examples how they contribute.

Q7. The crystal field splitting energy for $[\text{CoCl}_6]^{4-}$ is 18000 cm^{-1} . Calculate the crystal field splitting energy for $[\text{CoCl}_4]^{2-}$. [Atomic number of Co is 27]

Q8. The complex $[\text{Ni}(\text{H}_2\text{O})_6]^{2+}$ is green. If this is treated with excess ethylenediamine (en), forming $[\text{Ni}(\text{en})_3]^{2+}$, what happens to the color of the solution? [Atomic number of Ni is 28]